MICROBIOLOGY

Staphylococcus saprophyticus & Enterococcus

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GUE BLOCK

Microbiology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What single antibiotic susceptibility test distinguishes **S. saprophyticus** from other coagulase-negative staphylococci?
2. Why does **cephalosporin monotherapy** fail against Enterococcus, and what do **VanA/VanB** mechanisms change to confer vancomycin resistance?
3. A hospitalized patient with a catheter has an asymptomatic urine culture growing **Enterococcus faecium** — does this need antibiotics?

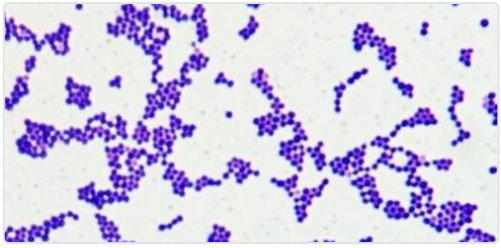
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Clinical importance — overview

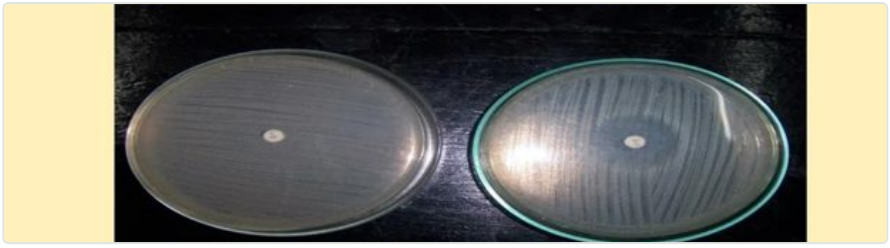
Feature	S. saprophyticus	Enterococcus (E. faecalis/faecium)
Key character	**Coagulase-negative** staphylococcus	Normal **intestinal** colonizer
Occurrence	Classically young, **sexually active women**	Important **opportunistic hospital** pathogen
Important disease	Major cause of **uncomplicated cystitis**	UTI, bloodstream infection, **endocarditis**

2 Staphylococcus saprophyticus

Morphology	Gram-positive cocci in **clusters**; **catalase positive**, **coagulase negative**; facultative anaerobe, nonmotile; usually **nonhemolytic**; naturally **resistant to novobiocin** — the key presumptive ID feature
Reservoir	Colonizes the **perineal/genitourinary** region (may occur in GI tract); often described as the **2nd commonest cause of UTI after E. coli**; infection usually follows ascent from periurethral flora
Virulence	**Surface adhesins** (attachment to uroepithelial cells) · **biofilm formation** (persistence on surfaces in some infections)
Clinical	Acute **cystitis** — dysuria, urgency, frequency, suprapubic discomfort ± hematuria. Uncomplicated cystitis: **usually no fever or flank pain** — fever/rigors/flank pain suggest upper/systemic UTI
Diagnosis	Clinical picture + pyuria/leukocyte esterase; urine **nitrite is often negative** (does not typically reduce nitrate — unlike the Enterobacterales); culture when indicated, confirm ID/AST by validated methods
Treatment	**Nitrofurantoin** is a common first-line option for localized uncomplicated cystitis; adjust for pregnancy/renal function/allergy/complicated disease. **Must not** be used for pyelonephritis or bacteremia



S. saprophyticus — Gram-positive cocci in clusters



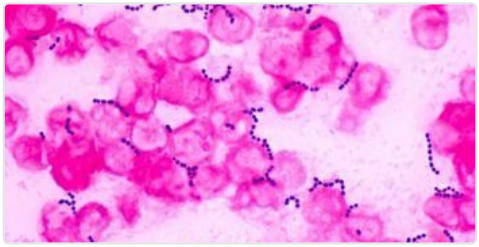
Novobiocin disc test — S. saprophyticus (resistant, left) vs S. epidermidis (sensitive, right)

Lab identification ladder

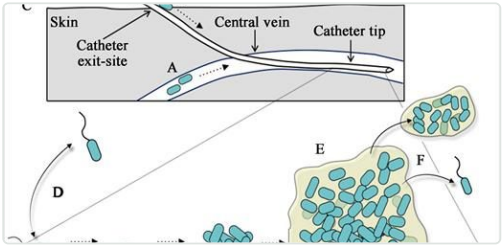
Test	Typical finding	Interpretation
Gram stain	Positive cocci in clusters	Suggests staphylococci
Catalase	Positive	Supports genus identification
Tube coagulase	Negative	Coagulase-negative staphylococcus
Novobiocin	**Resistant**	Supports *S. saprophyticus*
Vitek 2 / validated system	Species ID	Confirms clinically significant isolates

3 Enterococcus (E. faecalis, E. faecium)

Morphology	Gram-positive cocci, **pairs or short chains**; facultative anaerobe; **catalase negative** in routine ID; part of normal human **intestinal** flora; historically classified with group D streptococci
Reservoir	GI tract is the main reservoir — many infections arise from the **patient's own flora**; healthcare spread via contaminated hands/equipment/surfaces; risk ↑ with urinary catheters, prolonged admission, antibiotic exposure, impaired host defenses
Virulence	**Adhesins** (attachment to tissue/devices) · **biofilm** (persistence on catheters/prosthetic material) · some strains produce **cytolysin and gelatinase** · environmental tolerance aids survival in healthcare settings
Disease	**UTI** (esp. with urinary devices) · bloodstream infection incl. catheter-related bacteremia · **infective endocarditis** (esp. E. faecalis) · intra-abdominal/pelvic/postoperative wound infection · often healthcare-associated, particularly **E. faecium**
Why hard to treat	Intrinsic resistance → **cephalosporin monotherapy ineffective**; low-level aminoglycoside resistance limits monotherapy; **acquired high-level aminoglycoside resistance abolishes synergistic killing**; altered PBPs can cause ampicillin resistance
Treatment	Decide: symptomatic cystitis, systemic UTI, or colonization? Localized cystitis → active urinary drug by susceptibility (**amoxicillin or nitrofurantoin** in selected susceptible cases). Severe/ systemic disease → systemic-active therapy guided by AST + specialist advice. **Avoid treating asymptomatic bacteriuria** outside guideline exceptions



Enterococcus — Gram-positive cocci in pairs/short chains



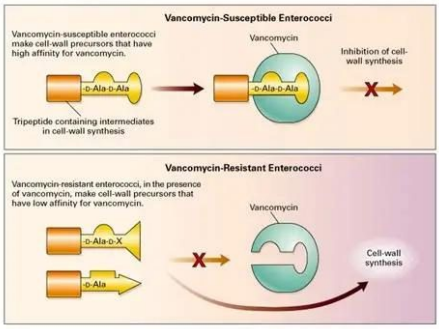
Biofilm formation on a catheter — a key route of persistence



Non-hemolytic E. faecalis colonies on blood agar

Acquired resistance & VRE

Vancomycin-resistant Enterococcus (VRE) is a major hospital threat; **E. faecium** carries a particularly high burden of multidrug resistance. **VanA/VanB mechanisms:** replace the usual **D-Alanine–D-Alanine (D-Ala-D-Ala)** peptidoglycan precursor — vancomycin's binding target — with the abnormal **D-Alanine–D-lactate (D-Ala-D-Lac)**, to which vancomycin cannot bind, blocking inhibition of cell-wall synthesis.



Vancomycin-susceptible vs -resistant Enterococcus — D-Ala-D-Ala vs D-Ala-D-Lac

Diagnosis

Test	Finding
Gram stain	Gram-positive cocci in pairs or chains
Catalase	Usually negative
Culture	Blood agar — mostly non-hemolytic
Identification	Species ID by Vitek 2 or validated system
Antibiotic sensitivities	Assess **vancomycin** and **high-level aminoglycoside resistance** when relevant

P Must-remember pearls

- **Novobiocin-resistant** coagulase-negative staphylococcus from a young, sexually active woman = **S. saprophyticus** cystitis.
- S. saprophyticus often gives a **negative urine nitrite** — unlike the nitrate-reducing Enterobacterales (E. coli, Klebsiella, etc.) — don't rule out infection on nitrite alone.
- **Cephalosporins never work** against Enterococcus — it is intrinsically resistant; don't reach for them empirically.
- **VanA/VanB** = D-Ala-D-Ala → D-Ala-D-Lac swap in the peptidoglycan precursor, removing vancomycin's binding target.
- **Asymptomatic bacteriuria with Enterococcus** (e.g. in a catheterized patient) is usually **not** treated with antibiotics — think catheter review, not culture-chasing.

Q Test yourself

1. A 22-year-old sexually active woman has dysuria, frequency and suprapubic pain, afebrile with no flank pain. Urine culture yields Gram-positive cocci in clusters, coagulase negative and novobiocin resistant. What is the most likely organism and first-line treatment?

A. Enterococcus faecalis; vancomycin
B. S. saprophyticus; nitrofurantoin
C. S. aureus; cephalosporin monotherapy
D. S. epidermidis; amoxicillin
E. Enterococcus faecium; cefepime

2. A 70-year-old hospitalized patient with an indwelling catheter has a urine culture growing Enterococcus faecium. He is afebrile with no urinary symptoms. What is the most appropriate management?

A. Start vancomycin immediately
B. Start amoxicillin for 7 days
C. No antibiotics — assess the catheter, this is likely asymptomatic bacteriuria/colonization
D. Start a cephalosporin
E. Start high-dose gentamicin monotherapy

3. Why is cephalosporin monotherapy never appropriate for a confirmed Enterococcus infection?

A. Enterococcus is always vancomycin-resistant
B. Enterococcus has intrinsic resistance to cephalosporins
C. Cephalosporins are nephrotoxic in all patients
D. Enterococcus lacks a cell wall
E. Cephalosporins only treat Gram-negative organisms

4. A hospital lab reports a vancomycin-resistant Enterococcus faecium (VRE) isolate. Which mechanism explains the resistance?

A. Efflux pump removing vancomycin from the cell
B. Altered penicillin-binding proteins only
C. Replacement of D-Ala-D-Ala with D-Ala-D-Lac in the peptidoglycan precursor, removing vancomycin's binding target
D. Beta-lactamase destruction of vancomycin
E. Loss of the bacterial cell wall entirely

5. A urine culture from a young woman grows Gram-positive cocci in clusters that are catalase positive, coagulase negative, and sensitive to novobiocin. What does the novobiocin sensitivity suggest?

A. Confirms S. saprophyticus
B. Argues against S. saprophyticus — consider another coagulase-negative staphylococcus (e.g. S. epidermidis)
C. Confirms Enterococcus
D. Confirms S. aureus
E. Is irrelevant to species identification

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — Novobiocin resistance in a coagulase-negative, catalase-positive staphylococcus from a young sexually active woman is the classic S. saprophyticus profile; nitrofurantoin is first-line for uncomplicated cystitis.

2 — Asymptomatic bacteriuria in a catheterized, afebrile patient without symptoms is usually not treated with antibiotics outside guideline exceptions — catheter assessment is the more appropriate step.

3 — Enterococcus has intrinsic resistance that makes cephalosporin monotherapy ineffective — this is a standing treatment principle, not isolate-specific.

4 — VanA/VanB mechanisms replace the normal D-Ala-D-Ala peptidoglycan precursor with D-Ala-D-Lac, to which vancomycin cannot bind, abolishing inhibition of cell-wall synthesis.

5 — Novobiocin resistance supports S. saprophyticus; a novobiocin-sensitive coagulase-negative staphylococcus argues against it, pointing instead to another CoNS species such as S. epidermidis.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam